

- Description

This module manages a smart gas water heater system, ensuring that:

1. The system continuously monitors the current water temperature (in °C).
2. If the temperature falls below the minimum threshold, the gas burner is activated to heat the water.
3. If the temperature exceeds the maximum threshold, the burner is turned off to prevent overheating.
4. If the temperature is within the optimal range, the system maintains the current state.

This system helps maintain an optimal water temperature, ensuring safety and energy efficiency.

- Specification

module SmartGasWaterHeaterControl

var currentTemperature, minTemperature, maxTemperature: real

var burnerActive: boolean

process updateTemperature(newTemperature: real) returns (currentTemperature: real, burnerActive: boolean)

functional scenario form:

currentTemperature < minTemperature && burnerActive = true, Burner activated.

||

currentTemperature > maxTemperature && burnerActive = false, Burner deactivated.

||

currentTemperature <= maxTemperature && burnerActive = false, Burner remains off.

end process

end module

- Java Code

```
package code;
public class c81_SmartGasWaterHeater {
    private double minTemperature;
    private double maxTemperature;
    private double currentTemperature;
    private boolean burnerActive;
    public c81_SmartGasWaterHeater(double minTemperature, double maxTemperature, double initialTemperature) {
        this.minTemperature = minTemperature;
        this.maxTemperature = maxTemperature;
        this.currentTemperature = initialTemperature;
        this.burnerActive = false;
    }
}
```

```

    }
    public void updateTemperature(double newTemperature) {
        currentTemperature = newTemperature;
        adjustBurnerState();
        System.out.printf("Current Water Temperature: %.2f°C, Burner: %
s%n",
                        currentTemperature,
                        burnerActive ? "ON" : "OFF");
        if (burnerActive) {
            System.out.println("ALERT: Low temperature detected. Burner
activated.");
        } else {
            System.out.println("Water temperature is optimal or too hig
h. Burner deactivated.");
        }
    }
    private void adjustBurnerState() {
        if (currentTemperature < minTemperature) {
            burnerActive = true;
        } else if (currentTemperature > maxTemperature) {
            burnerActive = false;
        }
    }
    public double getCurrentTemperature() {
        return currentTemperature;
    }
    public static void main(String[] args) {
        c81_SmartGasWaterHeater heater = new c81_SmartGasWaterHeater(40.0,
60.0, 50.0);
        // testcase-VT:
        //     heater.updateTemperature(35.0);
        //     heater.updateTemperature(65.0);
        //     heater.updateTemperature(50.0);

        // testcase-FT:
        //     heater.updateTemperature(-1.7976931348623157E308);
        //     heater.updateTemperature(1.7976931348614663E308);
        //     heater.updateTemperature(0.0);

        // testcase-VT:
        //     heater.updateTemperature(35.0);
        //     heater.updateTemperature(65.0);
        //     heater.updateTemperature(50.0);

```

```

// testcase-UVT:
heater.updateTemperature(35.0);
heater.updateTemperature(65.0);
heater.updateTemperature(50.0);
}
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c81_SmartGasWaterHeater.java	100.0 %	8	0	8
▼ c81_SmartGasWaterHeater	100.0 %	8	0	8
main(String[])		0	0	0
c81_SmartGasWaterHeater(doul		0	0	0
adjustBurnerState()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,050	21,139
▼ src	0.4 %	89	21,050	21,139
▼ code	0.4 %	89	21,050	21,139
▼ c81_SmartGasWaterHeater.java	96.7 %	89	3	92
▼ c81_SmartGasWaterHeater	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c81_SmartGasWaterHeater(doul	100.0 %	15	0	15
adjustBurnerState()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

4. Fuzz testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c81_SmartGasWaterHeater.java	87.5 %	7	1	8
▼ c81_SmartGasWaterHeater	87.5 %	7	1	8
main(String[])		0	0	0
c81_SmartGasWaterHeater(doul		0	0	0
adjustBurnerState()	75.0 %	3	1	4
updateTemperature(double)	100.0 %	4	0	4

5. Fuzz testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,050	21,139
▼ src	0.4 %	89	21,050	21,139
▼ code	0.4 %	89	21,050	21,139
▼ c81_SmartGasWaterHeater.java	96.7 %	89	3	92
▼ c81_SmartGasWaterHeater	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c81_SmartGasWaterHeater(doul	100.0 %	15	0	15
adjustBurnerState()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c81_SmartGasWaterHeater.java	100.0 %	8	0	8
▼ c81_SmartGasWaterHeater	100.0 %	8	0	8
main(String[])		0	0	0
c81_SmartGasWaterHeater(doul		0	0	0
adjustBurnerState()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,050	21,139
▼ src	0.4 %	89	21,050	21,139
▼ code	0.4 %	89	21,050	21,139
▼ c81_SmartGasWaterHeater.java	96.7 %	89	3	92
▼ c81_SmartGasWaterHeater	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c81_SmartGasWaterHeater(doul	100.0 %	15	0	15
adjustBurnerState()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c81_SmartGasWaterHeater.java	100.0 %	8	0	8
▼ c81_SmartGasWaterHeater	100.0 %	8	0	8
main(String[])		0	0	0
c81_SmartGasWaterHeater(doul		0	0	0
adjustBurnerState()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,050	21,139
▼ src	0.4 %	89	21,050	21,139
▼ code	0.4 %	89	21,050	21,139
▼ c81_SmartGasWaterHeater.java	96.7 %	89	3	92
▼ c81_SmartGasWaterHeater	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c81_SmartGasWaterHeater(doul	100.0 %	15	0	15
adjustBurnerState()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37